



Nursing Practice Guideline

Diabetes Patient Management

Reference #: NPG2

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Diabetes Patient Management. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

Introduction	1
Risk Statement	1
1.0 Principles	1
2.0 Blood Glucose Monitoring.....	2
3.0 Ketone Monitoring.....	3
4.0 Hypoglycaemia	5
5.0 Insulin Subcutaneous Injections	6
6.0 Diabetes Technologies - Continuous Subcutaneous Insulin Infusions – (Insulin Pumps) – Inpatient Management.....	8
7.0 Diabetes Technologies - Flash Glucose and Continuous Glucose Monitoring (Glucose sensors)	9
Related Documents	10
Authorisation & Version Control.....	12

1.0 Principles

Observe the five moments for hand hygiene. Follow aseptic technique during blood glucose monitoring procedure and use the risk assessment outlined in the SCGOPHCG Infection

Prevention and Management Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.

All prescriptions written within SCGOPHCG are to meet the minimum requirements set under the Medicines and Poisons Act 2014 and Medicines and Poisons Regulations 2016, in addition to best practice requirements.

2.0 Blood Glucose Monitoring

2.1 Safety Information

If the blood glucose meter is not calibrated it may result in inaccurate results¹

Freestyle Optium H Blood Glucose Monitor control testing using high/low control solution should be performed:

- Every 24 hours in the acute care setting
- When troubleshooting of questionable results
- If the monitor is dropped accidentally
- When the battery has been changed
- When opening a new box of test strips ¹

The blood glucose monitor must be cleaned after each use.

2.2 Nursing Management

- Routine Blood Glucose Level (BGL) testing **for patients on insulin**: pre meals, 2100 and 0200 hours.
Testing frequency may need to be increased related to patients' condition as indicated on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (Monitoring /Notification Instruction section)
- Routine Blood Glucose Level (BGL) testing **for patients not on insulin**:
 - Acute setting (SCGH and OPH Surgical Services): pre meals and 2100hours
 - Rehabilitation and Aged Care (OPH): QID for 48hours and then BD, or BD biweekly if stable.
- Document BGL results in the Insulin Subcutaneous Order and Blood Glucose Record, follow alert guidelines to notify Doctor as indicated.
- Continuous Glucose Monitors (Sensors) should NOT be used to replace Blood Glucose monitoring in the acute setting^{15,16}. There is insufficient data to support safe prescribing.

Steroids - Glucocorticoids

Patients' commencing/increasing or ceasing steroids require QID BGL for 48 hours and if the fasting BGL is >6mmol/L or a random BGL >10mmol/L BGL monitoring should be continued, and doctor notified to consider patient for therapeutic intervention³

- SCGH Replacement Blood Glucose monitors and logbooks can be obtained through Hospital Equipment Service, G block, Lower Ground. Ext 71482.
- OPH contact the CNS Diabetes
- To problem solve the monitor use the manufacturer's instructions as per the Freestyle Optium H Users Guide.¹

2.3 Procedural Information

Requirements

- Personal Protection Equipment (PPE)
- Blood Glucose Monitor (Freestyle Optium H Blood Glucose Monitor)
- Test Strip
- Safety Lancet
- Tissues / gauze
- Sharps Container

Procedure

1. Wash and dry patient's hands or wipe with a warm wet hand towel.
2. Perform Hand Hygiene.
3. Don PPE.
4. Open test strip packet by having the Blue side facing you tearing diagonally at the notch on the right-hand side in the foil. This will expose the electrode contact bars facing upward ready for insertion into the monitor.
5. Without turning on monitor, insert test strip. You have 3 minutes to apply blood.
6. Check that the lot number of the test strip appears on the screen, then when the blood drop symbol is displayed this indicates that a sample can be applied
7. Prepare the safety lancet (twisting off cap).
8. Obtain capillary blood sample from the lateral aspect of the patient's fingertip.
9. Apply blood to the test strip by bringing the blood drop to the white area at the end of the test strip. Blood is drawn into the test strip by capillary action.
10. Hold the blood drop to the test strip until 3 short lines are seen on the meter screen (this means enough blood is applied for a test).
11. If the countdown does not start, insufficient blood may have been applied, repeat test with new strip.
12. At completion of testing, apply gauze / tissue to fingertip.
13. Remove test strip (meter will turn itself off).
14. Dispose of safety lancet into sharps container. Perform hand hygiene.
15. Record result on Insulin Subcutaneous Order and Blood Glucose Record – Adult.
16. Clean monitor after each use.

3.0 Ketone Monitoring

3.1 Medical Requirements

If ketoacidosis is suspected, medical staff to consider checking venous blood gases, urea electrolytes (U&E), prescription of supplemental insulin or insulin infusion. Refer to SCGOPHCG Medication Management Guideline Adult Diabetic Ketoacidosis Guideline

3.2 Safety Information

Use of SGLT2i (Gliflozins) carries a risk of severe euglycaemic diabetes ketoacidosis (DKA). The risk is increased if the patient is fasting or has a restricted dietary intake, undergone a surgical procedure, is dehydrated or has an active infection. Please refer to SCGH Medication Management Guideline SGLT2 inhibitors (Gliflozins) and euglycaemic ketoacidosis for further information.

- Diabetic Ketoacidosis (DKA) is a life threatening, acute metabolic complication of Diabetes⁴
- Blood ketone testing enables rapid and accurate monitoring of ketone bodies⁵

- Urine ketone testing is no longer advocated in the acute care setting⁶

3.3 Nursing Management

Blood Ketone Testing should be initiated for:

- All patient with newly diagnosed Diabetes with BGL >15mmol/L
- All Type 1 patient (on admission/presentation to ED/persistent hyperglycaemia)
- All presenting patients receiving two or more insulin injections a day
- All patients on SGLT2i - on admission, fasting or limited oral intake
- Patients with diabetes who have symptoms of DKA (e.g. nausea, vomiting, abdominal cramps, drowsiness, hyperventilation, dehydration)
- As ordered by Doctor⁷

Interpretation of blood ketones using Freestyle Optium H Monitor^{1,7}

Blood Ketone value	Interpretation	Action
< 0.6 mmol/L	Normal	Treat Glucose level as per medical orders
0.6 – 1.0	Ketoacidosis potential	Notify Medical officer Re-check Ketones every 4 hours until blood ketone value <0.6 If tolerating diet and fluids encourage intake hourly
1.1 – 1.5 mmol/L	Ketoacidosis potential	Notify Medical officer Recheck ketones every 2 hours. If tolerating diet and fluids, encourage intake
> 1.5 mmol/L	Ketoacidosis probable	Notify Medical officer Recheck Ketones hourly Refer to Adult Diabetic Ketoacidosis Guideline (HRM025) **

**DKA guidelines can be accessed via SCGOPHCG HealthPoint

https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/SCGH/SCGOPHCG.MMG.Diabetic_Ketoacidosis_DKA.pdf

Requirements

- Personal Protection Equipment (PPE)
- Freestyle Optium H Blood Glucose Monitor
- Ketone Test Strip
- Safety Lancet
- Tissue/gauze
- Sharps container

Procedure

1. Wash and dry patient's hands or wipe with a warm wet hand towel
2. Perform hand hygiene and don PPE.
3. Open foil test strip by having the purple side facing you, tearing diagonally at the notch on the side. This will expose the contact bars to be inserted into the meter
4. Insert ketone test strip, this will turn on the meter (meter turns off after 3 minutes of inactivity).
5. 'KET' will appear on the screen.

6. Check that the lot number of the ketone test strip appears on the screen, then when the blood drop symbol is displayed this indicates that a sample can be applied
7. Prepare safety lancet (twisting off cap).
8. Obtain capillary blood sample from the lateral aspect of the patient's fingertip.
9. Apply blood to the test strip by bringing the blood drop to the white area at the end of the test strip, the blood is drawn into the test strip by capillary action
10. Hold the blood drop to the test strip until 3 short lines (- - -) are seen on the meter screen; this means enough blood is applied for a test.
11. If the countdown does not start, insufficient blood may have been applied - repeat test with new strip.
12. Apply tissue/gauze to patient's finger.
13. Remove test strip (meter will turn off itself).
14. Dispose of safety lancet into sharps container. Perform hand hygiene.
15. Clean monitor. Perform hand hygiene.
16. Document result on the Insulin Subcutaneous Order and Blood Glucose Record – Adult. Notify doctor if ketone level $>0.6\text{mmol/L}$.

4.0 Hypoglycaemia

4.1 Medical Requirements

Medical Officer should be informed of all hypoglycaemic incidents.

4.2 Safety Information

A patient is at an increased risk of hypoglycaemia if treated with insulin or insulin secretagogues (medication that stimulate endogenous insulin release) ⁸.

Hypoglycaemic unawareness (absence of neurogenic symptoms e.g. loss of early warning signs of hypoglycaemia) is common in patients with Type 1 Diabetes⁸.

Treat any $\text{BGL} < 4\text{mmol/L}$ with or without symptoms (8)

Refer to page 4 of Insulin Subcutaneous Order and Blood Glucose record- Adult. MR518 SCGH or MR156T OPH

Hypoglycaemia Management in Diabetes: BGL Less than 4mmol/L

4.3 Nursing Management

Follow the flowchart on page 4 of Insulin Subcutaneous Order and Blood Glucose record- Adult. MR518 SCGH or MR156T OPH

- Take into consideration patient's level of consciousness /NBM status/ presence of Insulin infusion.
- Flow Chart reflects principles of the 15g/15minute rule for Hypoglycaemia management.
- 15grams of fast Acting Carbohydrate i.e 60mls of glucose drink and rechecking BGL in 15 minutes.
- Followed up with 15 grams of slow acting carbohydrates when $\text{BGL} > 4\text{mmol/L}$

Ensure availability on each ward –

- Fast Acting Carbohydrate glucose drink (e.g., Carbotest or equivalent) and
- Slow Acting Carbohydrates e.g., biscuits or sandwiches

- Document blood glucose level on Insulin Subcutaneous Order and Blood Glucose Record – Adult MR 518(OPH)156T
- Notify Doctor of the event to review diabetes management and causes of hypoglycaemia.
- Correct avoidable causes and involve patient in discussion when appropriate to identify possible cause of hypoglycaemic event.
- Request the treating team to refer to the Endocrinology Team for advice if unable to identify the cause of hypoglycaemia to reduce the risk of further hypoglycaemic event.

5.0 Insulin Subcutaneous Injections

5.1 Medical Requirements

A valid prescription must be written by Medical Officer.

5.2 Safety Information

If the patient is fasting the insulin prescription must be reviewed by Medical Officer.

Timing of mealtime administration of insulin varies depending on the onset of action¹⁰

Rapid acting insulin (e.g. Novorapid/Humalog/Apidra/Fiasp) and premix Insulin containing rapid acting insulin (e.g. Novomix30/Humalogmix 25/Humalogmix 50/ Ryzodeg70/30) should be injected **immediately prior to meals**.¹⁰

Short acting insulin (e.g. Actrapid/Humulin R) and premix insulin containing short acting insulin (e.g. Mixtard 30/70/Humulin 30/70) should be injected **no more than 20-30 minutes prior to a meal**.¹⁰

Inpatients are to use Hospital Pharmacy dispensed insulin only

Patients own supply of insulin must be locked away with the patient's own medications on admission.

5.3 Nursing Management

- Insulin should be labelled with date and time opened- discard after 4 weeks. Patient addressograph label should be placed on pen injector device¹¹
- Each patient must have their own designated pen injector. Pen injector is kept in the locked medication drawer in the patient's room¹¹
- UNOPENED insulin should be stored in the dedicated medication refrigerator¹¹.
- Alcohol wipes must **not** be used to swab skin prior to injection¹¹
- Injection sites should be rotated to prevent lipohypertrophy or lipoatrophy. The preferred site for insulin injections is into the abdomen¹¹
- The patient should be assessed for suitability to administer own insulin using the prefilled insulin device (pen injector) with the BD Autosshield Duo™ safety pen needle and their suitability recorded (e.g. SCGH Diabetes Education Care Plan, OPH inpatient notes).
- BD Autosshield™ Safety Needle Patient Information Sheet is available and can be downloaded from Department of Endocrinology and Diabetes Insulin Safety Needle.
- Use the BD Autosshield Duo™ safety pen needle with the pen injector devices for **ALL** subcutaneous insulin administration by nursing staff and for patient self-administration

- Insulin is a high-risk medicine, two nurses are required to independently check and confirm the dosage and prescription prior to administration of subcutaneous Insulin.
- For patient self-administration of subcutaneous insulin, one nurse must first check the dose with the patient and observe administration.
- BD Autosshield Duo™ safety pen needles are single use only and must be disposed of in a sharp's container.

5.4 Procedural Information

PREFILLED INSULIN DEVICES - Pen injector (All Subcutaneous Insulin Administration)

Requirements

- Pen Injector
- 5mm BD Autosshield Duo™ safety pen needle
- Sharps container

Procedure

1. Perform hand hygiene.
2. Complete medication checks according to the 6 rights of medication administration.
3. Gently invert cloudy prefilled device 10 – 20 times (clear insulin preparations do not need to be inverted).
4. Attach BD Autosshield Duo™ safety pen needle.
5. Priming the pen injector:
 - a) For a new pen injector, dial up 4-6 units and expel into the air until insulin is visible at the end of the needle (if no insulin visible repeat step).
 - b) Prior to each subsequent injection dial up 2 units and expel into the air until insulin is visible at the end of the needle.
6. Dial up the prescribed insulin dose.
7. **Administration:**
 - a) **Patient self- administration:** nurse confirms the correct dose of insulin with patient.
 - b) **Nurse administration;** two nurses confirm the correct dose of insulin.
8. Using the palm (dagger) grip inject insulin into the subcutaneous tissue at a 90-degree angle. Once an audible click is heard (indicating engagement of the safety shield) depress dose selector until the dose selector returns to zero. Count to 10 then remove the device (Nb: skin fold lift technique is not required for routine injection however may be required for very thin patients to avoid intramuscular administration).
9. Remove BD Autosshield Duo Safety™ pen needle and dispose in a sharp's container.
10. Perform hand hygiene.
11. **Nurse** records administration on the Insulin Subcutaneous Order and Blood Glucose Record

Discharge Information

- The BD Autosshield Duo™ safety pen needle is for **HOSPITAL USE ONLY**

- Upon discharge instruct the patient to obtain the non-safety needles (size 4 to 6mm) for self-administration of insulin from a National Diabetes Service Scheme (NDSS) pharmacy.
- Patients commencing insulin need to register for subsidised/free supplies by completing the NDSS registration form (must be signed by a Medical Officer or Credentialed Diabetes Educator).
- For further information, including registration forms and locations of NDSS pharmacies see www.ndss.com.au
- For patients requiring Silver Chain support please provide patients with 7 days' supply of BD Autoschild Safety™ pen needles (As per Silver Chain Policy)

6.0 Diabetes Technologies - Continuous Subcutaneous Insulin Infusions – (Insulin Pumps) – Inpatient Management

6.1 Medical Requirements

Medical Officer will consult with Endocrinology Team as soon as possible.

Medical Officer must assess and document if the patient is deemed competent to continue insulin pump therapy as an inpatient. If the patient can demonstrate their physical and mental competency to manage their pump, they should be allowed to continue with pump therapy.

The assessment of competence will involve asking the patient to demonstrate the following:

- Open the management menu of their pump,
- Access their basal rates/ carbohydrate ratio/insulin sensitivity,
- Administer a bolus dose,
- Re-site their pump cannula (this could involve discussing how it is done and frequency of changes rather than observing the activity in the initial assessment)
- Technical competency regarding how they manage infusion line obstruction/site leaks,
- Adequate pump consumables e.g. infusion sets, reservoirs, batteries. Please note the hospital does not supply any insulin pump consumables.

If deemed competent and safe to self-manage, the Medical Officer will document the following:

- Brand of Pump,
- Type of insulin,
- Basal rates of insulin,
- Carbohydrate Ratio and Insulin Sensitivity settings.

6.2 Safety Information

If the insulin pump is to be discontinued the Medical Officer must prescribe subcutaneous insulin PRIOR to therapy being discontinued ¹².

Pumps should not be used in the setting of acute Diabetic Ketoacidosis (DKA) ¹²

Insulin pumps use fast acting (Novorapid/Humalog/Fiasp) insulin.

The insulin pump may need to be discontinued temporarily due to a number of circumstances during hospitalisation. Such circumstances where the insulin pump may need to be temporarily disconnected include:

- Surgical Procedures,
- Radiological investigation,
- CT Scan (pump must be removed),
- MRI scan (pump must be removed, including metal cannula infusion set),
- Physiotherapy (depending on the therapy),
- Hydrotherapy (if pump is not waterproof).

Discontinuation of the insulin pump for more than 30 minutes may result in significant hyperglycaemia, with the potential of Diabetic Ketoacidosis within 3 to 4 hours.

6.3 Nursing Management

Pumps have several functions, these include:

Bolus	a) Insulin given with meal based on blood glucose level, carbohydrate intake and physical activity. b) Insulin given to correct blood glucose levels out of target range. N.B Nursing staff to document amount of bolus insulin self-administered by the patient on the Insulin Subcutaneous Order and Blood Glucose record (M518) in the Administration Record section
Basal rate	Pre-programmed rates to deliver a continuous flow of insulin over 24 hours. Medical Officer to document in the integrated notes every 24 hours current basal rate settings and/or if any changes occur
Suspend/Stop	All insulin delivery stopped. Refer to safety information in 6.2

6.4 Monitoring

- Nursing staff to perform QID and 0200 Blood Glucose Level (BGL),
- Additional BGL may be undertaken at any time by the patient,
- Ketone testing as per Section. 3 -Ketone Monitoring,
- Current recommendation is for infusion set to be changed every 72 hours maximum or earlier as indicated by patient and documented by nursing staff in the inpatient notes.
- Nursing staff to check and document appearance of infusion site e.g. leakage, skin irritation and that the infusion line is not kinked.

7.0 Diabetes Technologies - Flash Glucose and Continuous Glucose Monitoring (Glucose sensors)

7.1 Safety Information

Both Flash Glucose Monitoring (FGM) and Continuous Glucose Monitoring Systems (CGMS) are **NOT** approved for inpatient glucose monitoring, they are designed for the ambulatory setting only. The information gathered from these devices should never be used to determine therapy changes.

Rationale - The level of glucose in interstitial fluid reacts slower than the level of glucose in the blood. This is because interstitial glucose results can lag 5 to 20 minutes behind the blood glucose result.

Inpatients may continue to wear their CGMS/FGM but Blood glucose monitoring using hospital supplied glucometer must continue as per section 2 – blood glucose monitoring. The BG results alone will be used to assess inpatient diabetes management and if any subsequent changes are required.

Contraindications for use of CGM/FGM in the inpatient setting

- Impaired level of consciousness
- Critical illness requiring intensive care
- Major psychiatric disturbance
- Refusal or unwillingness to participate in self-management
- Inability to demonstrate competency in managing their device
- Lack of equipment (the systems or consumables are not supplied by the hospital)
- Extensive skin infections or inflammation
- Any other medical circumstance deemed unsuitable by the treating medical or specialist endocrinology teams
- Radiological procedures involving strong magnetic or electromagnetic radiation i.e X-ray, MRI, CT or PET scan.

Nursing Management

7.2 Flash Glucose Monitoring

- Small wearable device that measures interstitial glucose levels
- Consists of a sensor (a fine electrode) inserted into the subcutaneous tissue on the back of the arm generally (exceptions sometimes apply)
- Sensor life 14 days
- Current FGM devices available in Australia: Freestyle Libre 2
- A reader and/or a smart phone app are required to scan over the sensor to see the current sensor glucose (SG) level. Reader or phone data capture provides the previous 8 hours of SG information. SG data can only be seen if there is regular scanning of the sensor.
- Flash glucose monitoring has optional alarms for high or low sensor glucose levels

7.3 Continuous Glucose Monitoring Systems (CGMS)

Enables continuous measurement of interstitial glucose levels. (N.B NOT Blood glucose levels)

Consist of 2 components:

- Disposable sensor electrode worn for 7 to 10 days (dependent on brand), that is inserted into the subcutaneous tissue as per patient preference and recommended placement area.
- Non disposable transmitter that is reused. Transmitters expire after 3 to 12 months dependent on brand
- CGMS data is communicated to either a smart phone app, insulin pump or handheld receiver
- The CGMS can be set with personalised alarms for high and low sensor glucose levels. Including the ability to suspend insulin delivery of a paired insulin pump if a low sensor glucose is either predicted or detected.
- Current CGMS devices available in Australia: Dexcom G6, Medtronic Guardian G3

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Policy:

- NMHS Clinical Documentation Policy
- SCGH Patient Identification and Procedure Matching Policy
- SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy
- WA Department of Health Mandatory Policy 0131/20 High Risk Medication Policy.
- Australian Commission on Safety and Quality in Health Care. National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines. Sydney: ACSQHC, 2015.

SCGOPHCG Medication Management Guidelines -

- Insulin: Adult Variable Rate Intravenous Insulin
- Adult Diabetic Ketoacidosis (DKA)
- Adult Hyperosmolar Hyperglycaemic State (HHS)
- SGLT2 inhibitors (Gliflozins) and euglycaemic ketoacidosis

SCGH Department of Endocrinology and Diabetes Guidelines

- Integrated Care Pathway for the management of Adults with Diabetes Mellitus at SCGH
- Complication Screening Management

Inpatient Guidelines -

- Peri-operative management guidelines
- Guidelines for patients undergoing colonoscopy
- Guidelines for patients on glucocorticoids
- Metformin and intravenous contrast
- Brittle Diabetes and Insulin Absorption Test

Relevant SCGOPHCG documentation includes:

- Adult Variable Rate Intravenous Insulin Management Record
- Adult Diabetic Ketoacidosis Guidelines and Management Record
- Adult Hyperosmolar Hyperglycaemic State Guidelines and Management Record

References

1. Abbott . Operators Manual. Freestyle Optium, blood Glucose and Ketone Monitoring System, 2013
2. Australian Diabetes Educators Association. ADEA Use of Blood Glucose Meters. Woden, ACT: ADEA March 2015
3. Swafe L, Narwani V, Stavrika C, Dhatariya K (2014): How frequently are bedside glucose measured in hospital inpatients on glucocorticoids? *Clinical Medicine* 14:327-8
4. Dunning T. Care of People with Diabetes 3rd ed. West Sussex. 2009 p 189
5. Guerci B, Tubiana-Rufi N, Bresson R, et al. Advantages to using capillary blood hydroxybutyrate determination for the detection and treatment of diabetic ketosis. *Diabetes Metabolism*. 2005 (31):401-406
6. Pluddenmann A, Heneghan C, Price CP, Wolstenholme J and Thompson M. Point –of-care blood test for Ketones in patients with diabetes: primary care diagnostic technology update. *British Journal of General Practice*. Aug 2011 : 530-531.
7. Australian Diabetes Educators Association. ADEA. Clinical Guiding Principles for Sick Day Management of Adults with Type1 and Type 2 Diabetes. Woden, ACT: ADEA May 2014
8. Endocrine Society. Evaluation and Mangement of Adult Hypoglycemic disorders. *Journal of Clinical Endocrinology and Metabolism*. Mar 2009;94(3):709-728
9. Insulin Subcutaneous Order and Blood Glucose Record. Sir Charles Gairdner Hospital. 2015
10. Cohen M. Handbook of Diabetes Management 8th ed. Caulfield, VIC: 2007
11. Australian Diabetes Educators Association. ADEA clinical recommendation: subcutaneous injection technique for insulin and glucagon like peptide. Woden, ACT: ADEA: 2011
12. Buchko BL, Artz B, Dayhoff S, March KS. Improving care of patients with insulin pumps during hospitalization: translating the evidence. *Journal of nursing care quality*. Oct 2012;27(4):330-340
13. Insulin infusion pump management. Inpatient Guidelines. Queensland Health Statewide Diabetes Clinical Network, July 2016
14. Australian diabetes Society Alert – Severe Euglycaemia ketoacidosis with SGLT2 inhibitor Use in the Perioperative Period
15. Gothong C, Singh LG, Satyarengga M, Spanakis EK. Continuous glucose monitoring in the hospital: an update in the era of COVID-19. *Curr Opin Endocrinol Diabetes Obes*. 2022 Feb 1;29(1):1-9.
16. Perez-Guzman MC, Shang T, Zhang JY, Jornsay D, Klonoff DC. Continuous Glucose Monitoring in the Hospital. *Endocrinol Metab (Seoul)*. 2021 Apr;36(2):240-255. doi: 10.3803/EnM.2021.201.

Authorisation & Version Control

Policy Sponsor	Nursing Practice Guideline Subcommittee				
Policy Contact	Nursing Practice Guideline Subcommittee Chair				
Date First Issued:	April 2001	Last Reviewed:	March 2023	Review Date:	March 2026
Version No.	8.0				
Clinical Reviewer:	J Kaye. Consultant Endocrinologist S Wilberforce CNC Diabetes N Cusack CN Diabetes				
Research Reviewer	S Wilberforce CNC Diabetes				
Executive Sponsor	M Walker, Deputy Nurse Co-director, NPGSc				
Endorsed by:	Nursing Practice Guideline Subcommittee			Date:	14/03/2023
Standards Applicable:	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.